Animal Breeding and Genetics

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Animal Breeding and Genetics (A32021)

Short Title:	Animal Breeding and Genetics		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Agriculture		
Author	KMURPHY		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module introduces the student to the fundamental concepts of molecular biology, genetics and genomics, and their applications to animal breeding. The module contains a practical component where the student can develop molecular laboratory skills, logical thinking, analysis and report-writing skills.

Programmes

AGRI-0106	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	3 / 5 / M

Indicative Content

- Overview of genetics and genomics
- Molecular genetics: structure and properties of nucleic acids, DNA replication and repair, transcription and translation, DNA extraction, PCR
- Domestication; qualitative and quantitative traits
- Crossbreeding, inbreeding
- Animal breeding and breeding policy
- National breed improvement strategies
- Genomic selection

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the fundamental concepts of molecular biology, genetics and genomics and their applications to animal breeding.
2. Demonstrate how molecular biology has improved the understanding of some areas of animal breeding.
3. Appraise the fundamental concepts and principles of animal breeding and demonstrate how these are applied to breed improvement strategies.
4. Communicate the importance of implementing breed improvement strategies.
5. Apply knowledge and skill to analyse livestock production systems and evaluate economic returns from such systems.
6. Use laboratory techniques and computer applications to demonstrate an understanding of molecular and quantitative genetics.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Guest industry speakers.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
Lab Report	20%	6
Project	20%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a good knowledge of the key learning outcomes including fundamentals of molecular and quantitative genetics, as well as a basic knowledge of animal breeding.

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of the learning outcomes, as well as being able to describe and evaluate national breed improvement strategies.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics, including factors affecting the rate of genetic gain and response to selection in breeding programmes and how this is applied to breed improvement strategies.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	24	
Independent Learning	75	

Essential Material

- Bray, D., K. Hopkin, B. Alberts, A. Johnson, M. Raff and Roberts, K & Walter, P . _Essential Cell Biology_. 3rd ed. USA: Garland Science, 2009.

Supplementary Material

- Campbell, N.A., J.B. Reece, L.A. Urry, M.L. Cain, S.A. Wasserman and Minorsky, P.V., Jackson, R.B. _Biology_. 8th ed. San Francisco: Pearson, Benjamin / Cummings, 2008.
- Simm, G. _Genetic Improvement of Cattle and Sheep_. UK: Farming Press, 1998.

Requested Resources

- Room Type: Computer Lab
- Lecture Room: Loose Seated
- Room Type: Biology Lab